

Kubernetes Operators in Java

Quarkus + JOSDK on OpenShift



QUARKUS



Who this is for

- Quarkus developers — REST, CDI, extensions, `quarkus:dev`
- Platform engineers curious about OLM and OperatorHub
- Integration folks who want Camel on OpenShift without reinventing the wheel

If you can write a `@ApplicationScoped` bean, you can write a Reconciler.

Agenda

Block	Content
<u>Theory</u>	Operators, JOSDK, CRD, reconcile, OLM bundle
<u>Live demo</u>	Community operator · <u>ephemeral only</u>

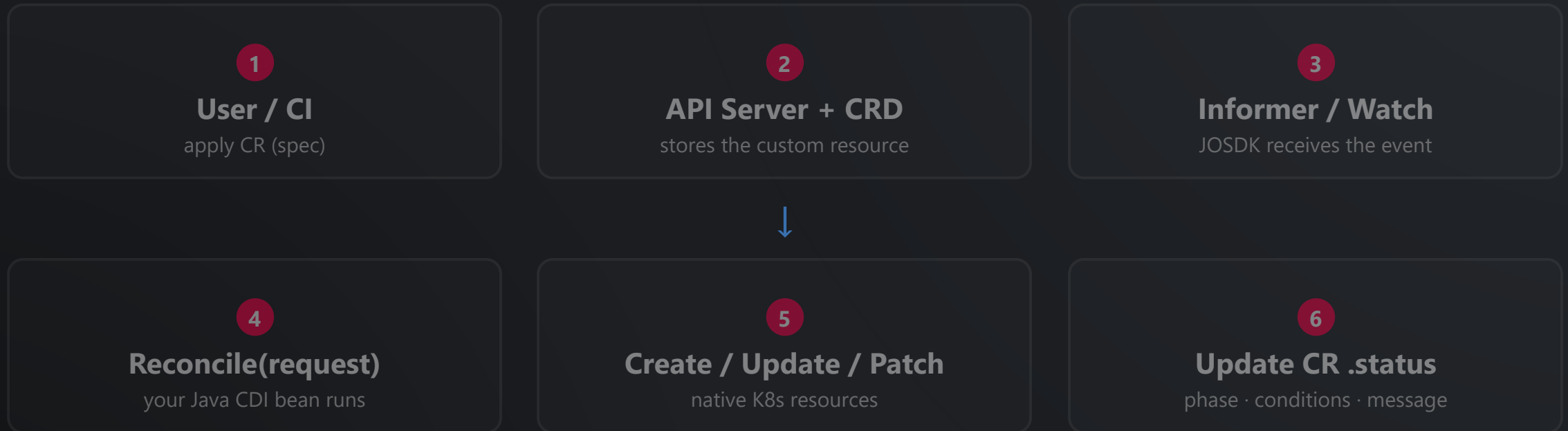
Why Kubernetes Operators?

Problem: Stateful apps on K8s need human ops — install, upgrade, backup, failovers.

Solution: Encode ops knowledge in software that runs **inside** the cluster.

- Extend the API with **Custom Resources (CRs)**
- A **controller** watches CRs and drives real resources (Deployments, Services, ...)
- **Declarative:** users set `spec`; operator converges cluster to desired state

Operator anatomy



Key idea: idempotent reconcile — safe to run again and again · loop repeats

The open-source stack



- **Quarkus** — fast startup, CDI, native image option
- **QOSDK** — Quarkus extension wrapping Java Operator SDK
- **Fabric8** — Kubernetes client (generated from CRDs)
- **OLM** — install / upgrade operators via catalog

You don't need Go

	Go (controller-runtime)	Java (QOSDK)
Ecosystem default	Yes	Growing
Team skills	Go required	<u>Java / Quarkus</u>
Dev experience	controller-gen	<u>code.quarkus.io</u> , <code>quarkus:dev</code>
OLM bundles	operator-sdk	<u>bundle-generator extension</u>

Quarkus Club: same toolchain you already use for microservices

Scaffold: code.quarkus.io + Quarkus CLI

1. Go to code.quarkus.io → search [qosdk](#) or [operator](#)
2. Select [Quarkus Operator SDK](#) extension
3. Optionally add [OLM Bundle Generator](#)
4. Generate & download — or:

```
quarkus create app org.acme:my-operator \
  --extension='quarkus-operator-sdk,quarkus-operator-sdk-bundle-generator'
cd my-operator
./mvnw quarkus:dev
```


Define your Custom Resource

Java types in `src/main/java/ ... /v1/` — JOSDK generates CRD YAML on build:

```
@Group("platform.io")
@Version("v1alpha1")
@Kind("IntegrationFlow")
public class IntegrationFlow extends CustomResource<
    IntegrationFlowSpec, IntegrationFlowStatus> {}
```

- **Spec** — desired state (user-facing)
- **Status** — observed state (operator-written)

Reference: [Joke sample](#)

The Reconciler — a CDI bean

```
@ControllerConfiguration(  
    csvMetadata = @CSVMetadata(  
        displayName = "Integration Operator",  
        bundleName = "openshift-integration-operator"))  
public class IntegrationFlowReconciler  
    implements Reconciler<IntegrationFlow> {  
  
    @Override  
    public UpdateControl<IntegrationFlow> reconcile(  
        IntegrationFlow resource, Context<IntegrationFlow> context) {  
        // compare spec → create/update Deployments, Services...  
        // update status phase: Pending → Running / Error  
        return UpdateControl.patchStatus(resource);  
    }  
}
```

Reconciliation loop · best practices

1. Level-triggered, not edge-only — reconcile on timer + watch
2. Idempotent — same spec → same outcome every time
3. Owner references — child resources garbage-collected with CR
4. Status conditions — Ready, Degraded, Progressing
5. Don't put secrets in spec — reference Secret names

Ephemeral mode in our demo uses ownerReferences so deleting the CR cleans up worker Deployments

OLM bundle generation

Add extension: `quarkus-operator-sdk-bundle-generator`

```
mvn clean package \
  -Dquarkus.container-image.build=true \
  -Dquarkus.container-image.registry=quay.io \
  -Dquarkus.container-image.group=maximilianopizarro \
  -Dquarkus.operator-sdk.bundle.channels=alpha
```

Output: `target/bundle/<name>/manifests/` + `bundle.Dockerfile`

Docs: [Deploy with OLM](#)

Deploy to OpenShift · test path

```
# Validate bundle

operator-sdk bundle validate target/bundle/openshift-integration-operator \
  --select-optional name=operatorhub

# Install from published bundle (community)

operator-sdk run bundle \
  quay.io/maximilianopizarro/openshift-integration-operator-bundle:v0.8.2 \
  --namespace openshift-integration \
  --install-mode AllNamespaces \
  --timeout 10m
```

Or: **OperatorHub** in OpenShift Console → Install

From skeleton to product

OpenShift Integration Operator — real-world QOSDK operator

- GitHub: [maximilianoPizarro/openshift-integration-operator](https://github.com/maximilianoPizarro/openshift-integration-operator)
- Apache 2.0 · OperatorHub · Quay.io
- **IntegrationFlow** CR — Camel routes, Kaoto design, ephemeral or GitOps
- Console dynamic plugin · 250+ flow templates

Today: **ephemeral Quick Try only** — GitOps is in the repo, not live today

Demo environment · try it at home

Today's path: [Podman Desktop](#) on RHEL + [OpenShift Local](#)

- Free [Red Hat Developer](#) subscription — sign in at [redhat.com](#)
- Full OpenShift console + OLM on your laptop — same path as the live demo
- Install guide: [developers.redhat.com/products/openshift-local](#)

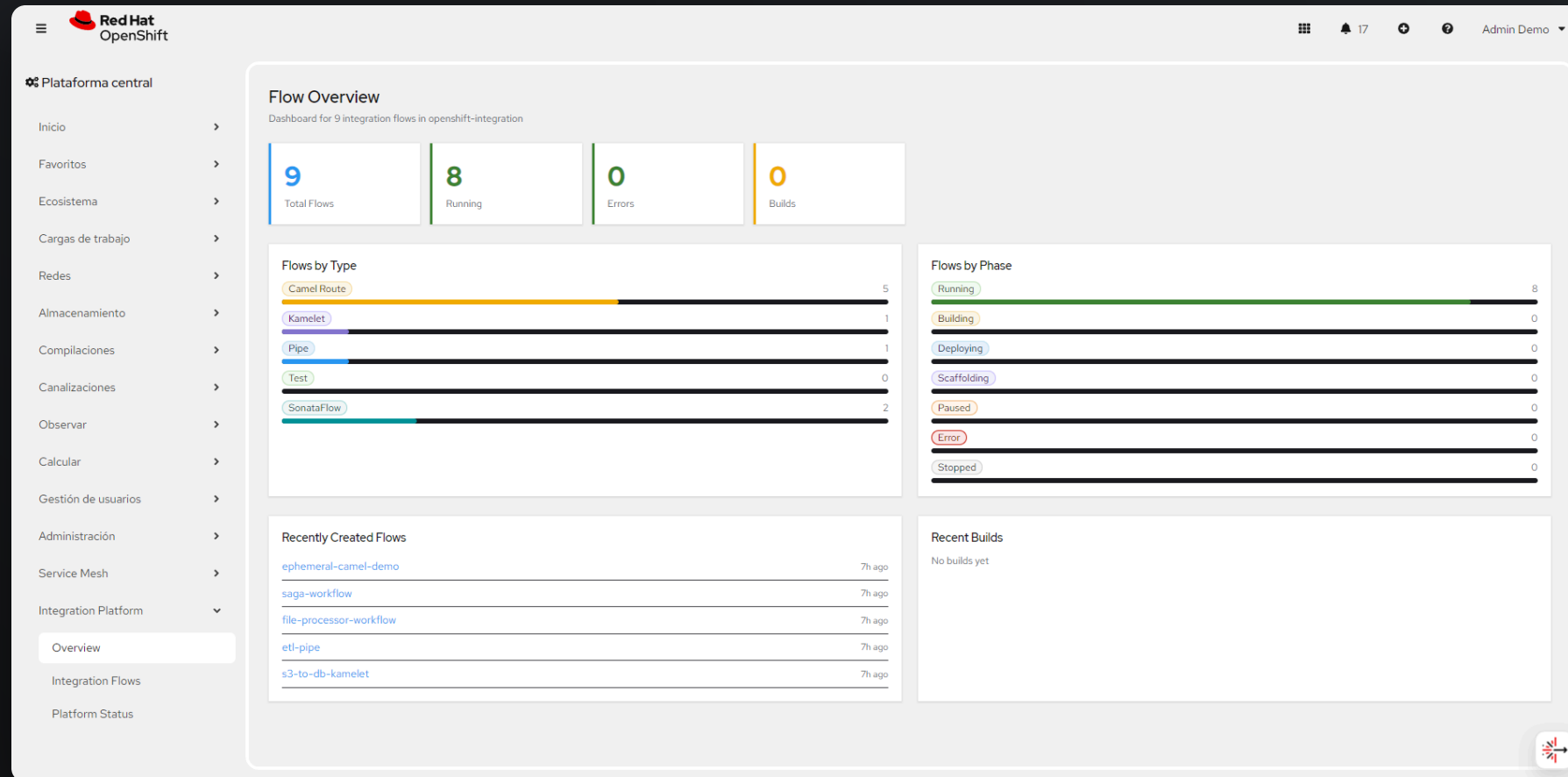
Prefer pure OSS? A [kind](#) cluster works for the operator runtime — you'll miss the OpenShift Console plugin UX.

Live demo

Ephemeral Quick Try on OpenShift Local

COMMUNITY OPERATOR · CONSOLE PLUGIN · KAOTO · NO GIT
REQUIRED

Demo · Install from OperatorHub



Operators → OperatorHub → "OpenShift Integration Operator" → Install

Demo · Console plugin

The screenshot displays the Red Hat OpenShift console interface. On the left, the 'Plataforma central' navigation menu is visible, with 'Integration Platform' selected. The main content area shows the 'Integration Flows' page, which lists 9 flows in the 'openshift-integration' namespace. The page includes filters for flow types (Camel Route, Kamelet, Pipe, SonataFlow) and phases (Expired, Running). A table lists the flows with columns for Name, Type, Mode, Phase, Clusters, Conditions, Repository, Resources, Age, and Actions.

Name	Type	Mode	Phase	Clusters	Condi...	Repository	Resources	Age	Actions
ephemeral-camel-demo	Camel Route	Ephemeral · expired	Expired	—	●●●	—	Logs Pods Pipelines	7h	Actions
error-handling-dlq	Camel Route	GitOps	Running	1/1	●●●	user/error-handling-dlq / @e48cb45	Logs Pods Builds Pipelines Argo	7h	Actions
etl-pipe	Pipe	GitOps	Running	1/1	●●●	user/etl-pipe / @f8936da	Logs Pods Builds Pipelines Argo	7h	Actions
file-processor-workflow	SonataFlow	GitOps	Running	1/1	●●●●	user/file-processor-workflow / @ad3fd37	Logs Pods Builds Pipelines Argo	7h	Actions
order-routing-cbr	Camel Route	GitOps	Running	1/1	●●●	user/order-routing-cbr / @f39ad77	Logs Pods Builds Pipelines Argo	7h	Actions
parallel-enrichment	Camel Route	GitOps	Running	1/1	●●●	user/parallel-enrichment / @585ae46	Logs Pods Builds Pipelines Argo	7h	Actions
rest-to-kafka	Camel Route	GitOps	Running	1/1	●●●	user/rest-to-kafka / @9548d1e	Logs Pods Builds Pipelines Argo	7h	Actions
s3-to-db-kamelet	Kamelet	GitOps	Running	1/1	●●●	user/s3-to-db-kamelet / @5c7b4b8	Logs Pods Builds Pipelines Argo	7h	Actions
saga-workflow	SonataFlow	GitOps	Running	1/1	●●●●	user/saga-workflow / @049a311	Logs Pods Builds Pipelines Argo	7h	Actions

Integration Platform in the OpenShift Console navigation

Demo · Create ephemeral IntegrationFlow

```
apiVersion: platform.io/v1alpha1
kind: IntegrationFlow
metadata:
  name: ephemeral-camel-demo
  namespace: openshift-integration
spec:
  deploymentMode: EPHEMERAL
  integrationType: CAMEL_ROUTE
  ephemeral:
    ttlSeconds: 3600
  kaotoDesign: |
    - route:
        from:
          uri: "timer:hello?period=5000"
          steps:
            - log: { message: "Hello from Quarkus Club!" }
```

Apply: `oc apply -f k8s/examples/09-ephemeral-demo.yaml`

Demo · Kaoto visual designer

The screenshot displays the Red Hat OpenShift console with the Kaoto visual designer embedded. The console header shows the Red Hat OpenShift logo and navigation icons. The left sidebar lists various system components like 'Plataforma central', 'Inicio', 'Favoritos', etc. The main area is divided into several sections:

- Flow Designer:** Shows a flow diagram for 'parallel-enrichment'. The flow starts with a 'platform-http' node, followed by a 'log' node, and then a 'multicast' node. The 'multicast' node branches into three 'direct' nodes, each leading to an 'Add step' button.
- YAML Editor:** Displays the current Kaoto design in YAML format. The design includes a route named 'parallel-enrichment' with a 'from' URI of 'platform-http://api/enrich' and a 'steps' section containing a 'log' step, a 'multicast' step, and another 'log' step.
- Node Telemetry:** A table showing the performance of the flow's components.
- Resources:** Links to various resources like Logs, Runtime Pods, Build Pods, PipelineRuns, ArgoCD App, and Git Repository.
- Target Clusters:** Shows the target clusters for the deployment.
- State:** Indicates the current state of the flow, which is 'running'.
- Last Commit:** Shows the last commit hash '585ae46'.
- Message:** Displays a message about the ApplicationSet reconciliation and cluster deployment.

Drag-and-drop Camel routes · embedded in the console plugin

Ephemeral architecture (today's path)

OpenShift Console (Plugin + Kaoto)

|

▼

IntegrationFlow CR (deploymentMode: EPHEMERAL)

|

▼

Operator Reconciler → EphemeralWorkerImageResolver

|

▼

Worker Deployment

|

▼

precompiled Camel Quarkus image (Quay)

|

▼

Logs / OTel / SSE → Console (Flow Logs, status)

GitOps (Gitea · Tekton · Argo CD) — in repo, not live today

Demo · Visual diagram & status

The screenshot displays the Red Hat OpenShift console interface. On the left, a sidebar menu lists various system components. The main area shows the 'saga-workflow' in a 'Running' state, with a 'GitOps' button and 'Pause', 'Stop', and 'Save Design' controls. Below this, a visual state machine diagram illustrates the workflow's logic, starting from 'InitState' and moving through 'CheckResult', 'SuccessState', 'ErrorState', 'processAction', 'notifySuccess', and 'handleError'. To the right of the diagram, a YAML editor shows the configuration for the workflow, including its ID, version, and state definitions. Further right, a 'Node Telemetry' table provides performance metrics for different components, and a 'Resources' section offers links to logs and other tools. The 'State' section indicates the workflow is 'running', and the 'Last Commit' is shown as '@49a311'.

Red Hat OpenShift

Plataforma central

- Inicio
- Favoritos
- Ecosistema
- Cargas de trabajo
- Redes
- Almacenamiento
- Compilaciones
- Canalizaciones
- Observar
- Calcular
- Gestión de usuarios
- Administración
- Service Mesh
- Integration Platform

← Back to Flows

saga-workflow Running GitOps

Engine: SONATAFLOW Branch: main

Pause Stop Save Design

YAML · LINES 7-8

```
1 id: saga-workflow
2 version: "1.0"
3 specVersion: "0.8"
4 name: Saga Workflow
5 start: InitState
6 states:
7   - name: InitState
8     type: operation
9   actions:
10     - functionRef:
11       refName: processAction
12     transition: CheckResult
13   - name: CheckResult
14     type: switch
15   dataConditions:
16     - condition: "${ .result == true }"
17     transition: SuccessState
18   defaultCondition:
19     transition: ErrorState
20   - name: SuccessState
21     type: operation
22   actions:
23     - functionRef:
24       refName: notifySuccess
25   end: true
26   - name: ErrorState
27     type: operation
28   actions:
29     - functionRef:
30       refName: handleError
31   end: true
32 functions:
33   - name: processAction
34     type: expression
35     operation: ".result = true"
```

Node Telemetry

http-source-1	85ms	1774rpm	●
log-processor	15ms	2584rpm	●
kafka-sink	34ms	305rpm	●
transform-1	177ms	3499rpm	●

Updated: 22:41:50

Resources

- Logs
- Runtime Pods
- Build Pods
- PipelineRuns
- ArgoCD App
- Git Repository
- SonataFlow Console
- SonataFlow CR

Target Clusters

None

State

running

Last Commit

@49a311

Message

ApplicationSet reconciled, tracking 1 cluster deployment(s)

Reconciler wrote status · diagram reflects route structure

Demo · Watch reconciliation

```
oc get integrationflow -n openshift-integration -w
```

```
oc get pods -n openshift-integration -l integrationflow=ephemeral-camel-demo
```

```
oc logs -f deploy/iflow-ephemeral-camel-demo-worker -n openshift-integration
```

Expect: **phase Running** · pod **2/2 or 1/1 Ready** · logs every 5s

Demo · Flow logs in console

The screenshot displays the Red Hat OpenShift console interface. On the left is a sidebar menu with options like 'Inicio', 'Favoritos', 'Ecosistema', 'Cargas de trabajo', 'Redes', 'Almacenamiento', 'Compilaciones', 'Canalizaciones', 'Observar', 'Calcular', 'Gestión de usuarios', 'Administración', 'Service Mesh', and 'Integration Platform'. The main area shows the 'demo-test' flow, which is currently 'Paused' and has an 'Ephemeral- 58m left' status. Below the flow name are tabs for 'Visual Flow', 'Kaoto Designer', 'YAML Editor', 'History', 'Dependencies', 'Spec', 'Logs', and 'Status'. The 'Logs' tab is active, showing a list of log entries with timestamps and messages. A 'Follow' button is visible next to the log list. On the right side of the console, there is a 'Node Telemetry' section showing metrics for various components (http-source-1, log-processor, kafka-sink, transform-1) and a 'Resources' section showing the 'local' cluster. The 'State' section shows the flow is 'paused'. The 'Expires At' section shows the flow will expire on 5/6/2026, 23:43:30. The 'Message' section shows the flow is 'Ephemeral flow paused by user'.

Red Hat OpenShift

Plataforma central

Inicio

Favoritos

Ecosistema

Cargas de trabajo

Redes

Almacenamiento

Compilaciones

Canalizaciones

Observar

Calcular

Gestión de usuarios

Administración

Service Mesh

Integration Platform

demo-test Paused Ephemeral- 58m left

Engine: CAMEL_ROUTE · Clusters: local

Visual Flow Kaoto Designer YAML Editor History Dependencies Spec Logs Status

500 lines Follow Refresh

openshift-integration/flow-demo-test-worker-77f648f5c9-7tn49 (worker) · Updated 22:44:42 · [Open in console](#)

```
-- / _ \ / / / / / _ \ / / / / /
- / / / / / _ \ / / / / /
- \ \ \ \ \ / / / / / /
-- \ \ \ \ \ / / / / / /

2026-06-06 01:44:38,993 INFO [org.apache.camel.quarkus.core.CamelBootstrapRecorder] (main) Apache Camel Quarkus 3.36.0 is starting
2026-06-06 01:44:38,996 INFO [org.apache.camel.main.MainSupport] (main) Apache Camel (Main) 4.20.0 is starting
2026-06-06 01:44:39,131 INFO [org.apache.camel.main.BaseMainSupport] (main) Auto-configuration summary
2026-06-06 01:44:39,132 INFO [org.apache.camel.main.BaseMainSupport] (main) [MicroProfilePropertiesSource] camel.main.routesIncludePattern = file:/deployments/config/flow.camel.yaml
2026-06-06 01:44:39,132 INFO [org.apache.camel.main.BaseMainSupport] (main) [MicroProfilePropertiesSource] camel.main.routesReloadEnabled = true
2026-06-06 01:44:39,253 INFO [org.apache.camel.impl.engine.AbstractCamelContext] (main) Apache Camel 4.20.0 (camel-1) is starting
2026-06-06 01:44:39,276 INFO [org.apache.camel.impl.engine.AbstractCamelContext] (main) Routes startup (total:1)
2026-06-06 01:44:39,276 INFO [org.apache.camel.impl.engine.AbstractCamelContext] (main) Started new-route (timer://tick)
2026-06-06 01:44:39,276 INFO [org.apache.camel.impl.engine.AbstractCamelContext] (main) Apache Camel 4.20.0 (camel-1) started in 22ms (build:0ms init:0ms start:22ms boot:493ms)
2026-06-06 01:44:39,282 INFO [io.quarkus] (main) camel-worker-core 0.3.0 on JVM (powered by Quarkus 3.36.1) started in 0.863s.
2026-06-06 01:44:39,282 INFO [io.quarkus] (main) Profile prod activated.
2026-06-06 01:44:39,282 INFO [io.quarkus] (main) Installed features: [camel-bean, camel-direct, camel-log, camel-seda, camel-timer, camel-yaml-dsl, cdi]
2026-06-06 01:44:40,292 INFO [new-route] (Camel (camel-1) thread #1 - timer://tick) Hello Integration Platform
2026-06-06 01:44:40,294 INFO [info] (Camel (camel-1) thread #1 - timer://tick) Exchange[ExchangePattern: InOnly, BodyType: null, Body: [Body is null]]
```

Node Telemetry

http-source-1	123ms	3399rpm	●
log-processor	68ms	3874rpm	●
kafka-sink	191ms	2183rpm	●
transform-1	42ms	1145rpm	●

Updated: 22:44:42

Resources

[Logs](#)

[Runtime Pods](#)

Target Clusters

local

State

paused

Ephemeral Worker

deployment/flow-demo-test-worker

Expires At

5/6/2026, 23:43:30

Message

Ephemeral flow paused by user

Live pod log streaming · container selector · follow mode

Demo · Platform status

The screenshot shows the Red Hat OpenShift Platform Status dashboard. The top navigation bar includes the Red Hat OpenShift logo, a hamburger menu, and user information (Admin Demo). The left sidebar lists navigation options under 'Plataforma central', with 'Platform Status' selected. The main content area is titled 'Platform Status' and shows '6/7 services healthy - Last checked: 22:39:04'. Below this, five summary cards display: 9 Integration Flows, 0 PipelineRuns, 209 ArgoCD ApplicationSets, 1 Ephemeral (Quick Try) with a 'Create Quick Try flow' link, and 6/7 Services Healthy. A 'Refresh' button is in the top right. The 'External Services' section lists several components with their health status: Integration Operator (HEALTHY, 1/1 replicas ready), Console Plugin (HEALTHY, 1/1 replicas), Kaoto Designer (HEALTHY, 1/1 replicas), Gitea (Git Server) (HEALTHY, 1 replicas in gitea namespace), ArgoCD (GitOps) (HEALTHY, Server: 1 replicas), Tekton Pipeline (ERROR, Pipeline not found), and OpenTelemetry Collector (HEALTHY, 1 replicas). A 'Platform Pods (19)' section is partially visible at the bottom. A small red alert icon is in the bottom right corner.

Service	Count	Status
Integration Flows	9	Healthy
PipelineRuns	0	Healthy
ArgoCD ApplicationSets	209	Healthy
Ephemeral (Quick Try)	1	Healthy
Services Healthy	6/7	Healthy

Service	Description	Health	Replicas
Integration Operator	Quarkus operator reconciling IntegrationFlow CRs	HEALTHY	1/1 replicas ready
Console Plugin	OpenShift Console Dynamic Plugin serving this UI	HEALTHY	1/1 replicas
Kaoto Designer	Visual editor for Apache Camel integrations	HEALTHY	1/1 replicas
Gitea (Git Server)	Hosts scaffolded worker source code repositories	HEALTHY	1 replicas in gitea namespace
ArgoCD (GitOps)	Deploys workers to target clusters via ApplicationSets	HEALTHY	Server: 1 replicas
Tekton Pipeline	CI pipeline for building worker container images	ERROR	Pipeline not found
OpenTelemetry Collector	Receives traces and metrics from worker pods	HEALTHY	1 replicas

Health: Operator · Kaoto · OTel · (GitOps services when enabled)

Demo · TTL & lifecycle

The screenshot displays the Red Hat OpenShift Platform Central interface. On the left is a sidebar with navigation links: Inicio, Favoritos, Ecosistema, Cargas de trabajo, Redes, Almacenamiento, Compilaciones, Canalizaciones, Observar, Calcular, Gestión de usuarios, Administración, Service Mesh, and Integration Platform. The main area shows the configuration for a flow named 'parallel-enrichment', which is in a 'Running' state. The configuration is shown in the 'Spec' tab, displaying a YAML snippet for a Camel route. The right sidebar provides telemetry and resource information.

parallel-enrichment Running GitOps

Engine: CAMEL_ROUTE · Branch: main

Visual Flow Kaoto Designer YAML Editor History Dependencies **Spec** Logs Status

```
{
  "branch": "main",
  "engine": "CAMEL",
  "gitRepository": "https://gitea.example.com/user1/parallel-enrichment",
  "integrationType": "CAMEL_ROUTE",
  "kaotoDesign": "- route:\n  id: parallel-enrichment\n  from:\n    uri: \"platform-http://api/enrich\"\n  steps:\n    - log:\n      message: \"Enriching: ${body}\"
  targeting: {
    \"clusters\": [
      \"local\"
    ],
    \"strategy\": \"explicit\"
  }
}
```

Node Telemetry

http-source-1	62ms	4768rpm	●
log-processor	19ms	2119rpm	●
kafka-sink	191ms	4601rpm	●
transform-1	114ms	346rpm	●

Updated: 22:46:24

Resources

- [Logs](#)
- [Runtime Pods](#)
- [Build Pods](#)
- [PipelineRuns](#)
- [ArgoCD App](#)
- [Git Repository](#)

Target Clusters

None

State

running

Last Commit

585ae46

Message

ApplicationSet reconciled, tracking 1 cluster deployment(s)

ttlSeconds · extend from console · optional promote-to-gitops (not live)

Try it yourself

Easiest path (what we demo)

- [OpenShift Local](#) via Podman Desktop — free Developer subscription
- Install community operator from [OperatorHub](#) · apply `09-ephemeral-demo.yaml`

Also fine: ROSA / ARO / on-prem OpenShift · or **kind** + `operator-sdk run bundle`
(no Console plugin)

Quick start: [docs site](#)

What you learned

1. Operators extend K8s with CRDs + reconcile loops
2. QOSDK scaffolds from code.quarkus.io — Java, not Go
3. Reconciler = CDI bean; bundle-generator → OLM → OperatorHub
4. Real operator: IntegrationFlow ephemeral mode on OpenShift
5. Community path: you can install and try today on your cluster

Links & resources

Resource	URL
OpenShift Local	https://developers.redhat.com/products/openshift-local
QOSDK docs	https://docs.quarkiverse.io/quarkus-operator-sdk/dev/
Joke sample	https://github.com/quarkiverse/quarkus-operator-sdk/tree/main/samples/joke
Integration Operator	https://github.com/maximilianoPizarro/openshift-integration-operator
Docs & Quick Start	https://maximilianopizarro.github.io/openshift-integration-operator/
OperatorHub	Search "OpenShift Integration Operator"
Quarkus Club	https://www.youtube.com/live/WwMOUMQyq4s
LinkedIn	https://www.linkedin.com/in/maximiliano-gregorio-pizarro-consultor-it/

Thank you

Quarkus Club

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